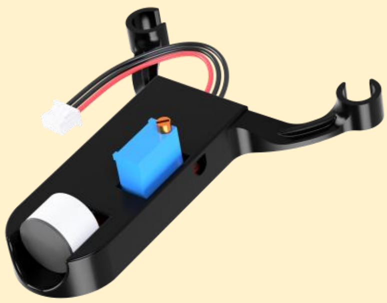
🔊 SOUND_WAKEUP 🚨

About :

Pluto drone is equipped with a sound sensor, utilized for detecting loud sounds. It is used to wake Up(take off) the drone when some loud sound (thud/bang) is detected. It can be used to automate drone operations

Add-On used :

A sound sensor module detects sound intensity in the surrounding environment. It does not recognize words or voices, but only measures how loud a sound is.



Components and Connection:

Pluto drone, Sound sensor module and connecting wires.

Connect the Sound sensor on the ADC port of Primus V5/X2 board

Primus V5/X2 pin	Peripherals
VCC on ADC port	VCC on Sound Sensor
GND on ADC port	GND on Sound Sensor
Signal on ADC port	A0 on Sound Sensor

How does it work?

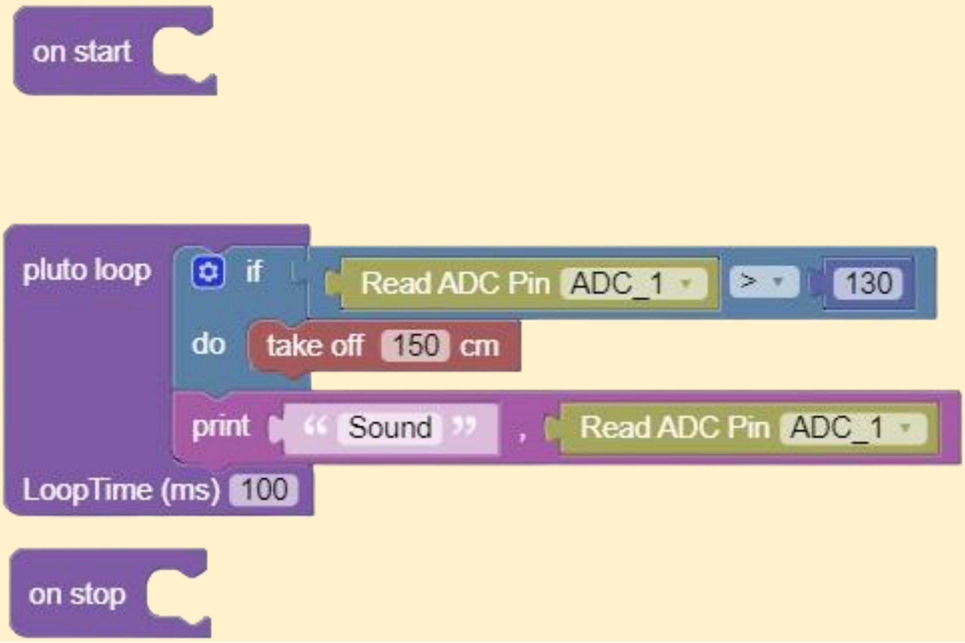
Think of the sound sensor like a loudness detector:

- **Normal reading → 110-125 = No_takeoff**
- **Loud Sound → 130+ reading = Drone Take_off**

The program continuously checks the sensor:

- If smoke is detected → **Buzzer turns ON**
- If no smoke → **Buzzer stays OFF**

PlutoBlocks



Working:

Initially, we set Reciever mode as ESP
When loop starts executing it constantly checks sound sensor readings whether it has sensed anylou sound. If it has, it takes off the drone and initiate normal flying.

Fig: Sound sensor add-on Pluto

Moving further

Try making adding a different type of sensor, which is more sensitive and responds differently to sound levels.